

OPT2

Mathematics Learning and Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a ninth-grade class that meets during first period, most students divide fractions correctly with “keep, change, flip” but cannot say why it works or solve a word problem that needs it. Their teacher has taught for 22 years. What should she do next?

- A.** Use visual models and “how many groups of ...?” contexts to connect division of fractions to its meaning.
- B.** Move on, since they get the right answers.
- C.** Assign more practice with the same rule until it is automatic.
- D.** Teach a different shortcut, such as common denominators.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A student explains clearly why the slope of a line is a constant rate of change but is slow and error-prone when calculating slope from two points. What is the best next step?

- A.** Move him to a lower-level course.
- B.** Re-teach the meaning of slope from the beginning.
- C.** Let him use a calculator for every slope problem from now on.
- D.** Give targeted practice with the slope formula, linked to the rate-of-change idea he already has.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Midway through a linear-equations unit, a teacher gives a 5-question exit ticket printed on half sheets. Two-thirds of the class misses the questions that involve distributing a negative sign. The unit test is in six days. What makes this exit ticket formative?

- A.** It is given before the unit test.
- B.** It is short and ungraded.
- C.** The teacher uses the results to re-teach distributing negatives before moving on.
- D.** It is recorded in the gradebook so students take it seriously.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A teacher wants to know, at the end of a semester, whether students met the course's algebra standards. Which kind of assessment fits that purpose?

- A.** A summative assessment aligned to the standards
- B.** An attitude survey about math
- C.** A formative exit ticket
- D.** A diagnostic pre-assessment

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On a warm-up projected on a new interactive board, 14 of 28 students write $(x + 3)^2 = x^2 + 9$. The class has a substitute on Fridays. What is the best teacher response?

- A.** Tell students to be more careful next time.
- B.** Wait until Friday so the substitute can review it.
- C.** Have students test $x = 1$ in both expressions, see $16 \neq 10$, and then use an area model to find the missing $6x$.
- D.** Restate the FOIL rule and assign 20 more problems.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Students in a class that uses a 1-to-1 laptop program can solve $y = 2x + 3$ for y when $x = 4$, but when shown the graph they cannot find where the line crosses $y = 11$. The school year started three weeks ago. Which activity best addresses this?

- A.** Skip graphs, since the equation always gives the answer.
- B.** Give more equation-solving practice.
- C.** Have students build a table from the equation, plot the points, and explain how each row matches a point on the line.
- D.** Show a graph and an equation on separate days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

During a 50-minute block, groups work on a rich task: finding how many handshakes happen when 20 people each shake hands once. Several groups are stuck after 10 minutes. The room's air conditioning is loud. What should the teacher do?

- A. Move all groups to an easier worksheet.
- B. Ask, "What happens with 3 or 4 people? What pattern do you notice?"
- C. Let them keep working without any help, since struggle is always good.
- D. Write $n(n - 1)/2$ on the board so they can finish.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A teacher's original task asked students to design and justify a fair way to split a shared phone bill. After revising, the task lists three steps and a fill-in formula. What happened to the task's cognitive demand?

- A. It dropped from a high-demand reasoning task to procedure practice.
- B. It rose, because a formula is more advanced mathematics.
- C. It stayed the same, since the context did not change.
- D. It rose, because there are now more steps.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A student in a class of 30 says the sum of two odd numbers is always odd. The lesson is being recorded for a mentoring program. Which teacher question is an advancing question that keeps the student reasoning?

- A. "Isn't $3 + 5$ equal to 8, and isn't 8 even?"
- B. "Can you test your claim with a few pairs and explain what you notice?"
- C. "What did you mean by 'odd'?"
- D. "That's wrong; the sum is always even."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A teacher asks, “Is the slope positive? So it goes up? So the answer is B, right?” What kind of questioning is this?

- A.** Wait time
- B.** Assessing, which finds out what the student thinks
- C.** Focusing, which keeps the student reasoning
- D.** Funneling, which leads the student to the answer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A new teacher with a well-stocked supply closet is planning a unit on exponential functions. She has found a fun candy-decay activity and a popular video. Using backward design, what should she decide first?

- A.** The final test, written before the goals
- B.** Where the candy activity and video fit in the unit
- C.** The textbook’s page order for the chapter
- D.** The learning goals students should reach by the end of the unit

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A student who moved from another state in October can set up a proportion but cannot solve it without help. His family recently adopted a puppy. What is the best support?

- A.** Keep the same supports in place all year.
- B.** Give a worked example and guiding prompts on the solving step, then fade them as he succeeds.
- C.** Assign simpler, non-proportion problems instead.
- D.** Solve the problems for him until he catches up.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

In a ninth-grade class ~~that meets during first period~~, most students divide fractions correctly with “keep, change, flip” but cannot say why it works or solve a word problem that needs it. ~~Their teacher has taught for 22 years.~~ What should she do next?

Target word: next instructional move **Decides:** The procedure works but meaning is missing.

A	Correct	Builds the concept the procedure rests on, which supports transfer to word problems.
B	Absolute language	Correct answers alone do not show understanding or transfer.
C	Dropped element	Drops meaning; the procedure is already automatic.
D	Adjacent category	Swaps one procedure for another without building meaning.

2. Answer D Concept 1 • transfer

A student explains clearly why the slope of a line is a constant rate of change but is slow and error-prone when calculating slope from two points. What is the best next step?

Target word: concept present, procedure weak — reverse case **Decides:** Understanding exists; fluency is missing.

A	Adjacent category	A narrow procedural gap does not call for a placement change.
B	Inverted operation	Treats a fluency gap as a concept gap.
C	Absolute language	Avoids building the fluency he needs.
D	Correct	Once the concept is in place, practice builds efficient procedure.

3. Answer C Concept 2

Midway through a linear-equations unit, a teacher gives a 5-question exit ticket ~~printed on half sheets~~. Two-thirds of the class misses the questions that involve distributing a negative sign. ~~The unit test is in six days.~~ What makes this exit ticket formative?

Target word: what makes it formative **Decides:** Evidence is used to change instruction while learning is still happening.

A	Dropped element	Timing alone is not enough; the evidence has to change what happens next.
B	Adjacent category	Length and grading do not make an assessment formative; use does.
C	Correct	Formative assessment is defined by the adjustment it drives.
D	Confusion set	Grading for the record is a summative purpose.

4. Answer A Concept 2 • transfer

A teacher wants to know, at the end of a semester, whether students met the course’s algebra standards. Which kind of assessment fits that purpose?

Target word: purpose → assessment type (reverse) **Decides:** End-of-course judgment of mastery is summative.

A	Correct	It evaluates learning after instruction.
B	Adjacent category	Measures dispositions, not mastery of standards.
C	Inverted operation	Formative checks guide instruction during learning.
D	Confusion set	Diagnostic tests measure readiness before instruction.

5. Answer C Concept 3

On a warm-up projected on a new interactive board, 14 of 28 students write $(x + 3)^2 = x^2 + 9$. The class has a substitute on Fridays. What is the best teacher response?

Target word: respond to a misconception **Decides:** Half the class shares one consistent error pattern.

A	Confusion set	Treats a systematic misconception as a careless slip.
B	Camouflage	The substitute schedule is irrelevant; the misconception needs attention now.
C	Correct	Cognitive conflict followed by a model addresses the belief behind the error.
D	Dropped element	Drops the student reasoning that produced the error.

6. Answer C Concept 4

Students in a class that uses a 1 to 1 laptop program can solve $y = 2x + 3$ for y when $x = 4$, but when shown the graph they cannot find where the line crosses $y = 11$. The school year started three weeks ago. Which activity best addresses this?

Target word: connect representations **Decides:** Students work in one representation but cannot link it to another.

A	Absolute language	Graphs are a key representation students must interpret.
B	Dropped element	Strengthens the form they already handle and leaves the graph out.
C	Correct	Explicitly linking table, graph, and equation builds connections.
D	Adjacent category	Adds representations without connecting them.

7. Answer B Concept 5

During a 50-minute block, groups work on a rich task: finding how many handshakes happen when 20 people each shake hands once. Several groups are stuck after 10 minutes. The room's air conditioning is loud. What should the teacher do?

Target word: support struggle without lowering demand **Decides:** Keep students doing the reasoning.

A	Adjacent category	Replaces the task instead of supporting it.
B	Correct	A question points toward a strategy without doing the thinking for them.
C	Absolute language	Productive struggle needs support; unsupported struggle can become frustration.
D	Inverted operation	Giving the formula turns a thinking task into computation.

8. Answer A Concept 5 • transfer

A teacher's original task asked students to design and justify a fair way to split a shared phone bill. After revising, the task lists three steps and a fill-in formula. What happened to the task's cognitive demand?

Target word: judge demand after a revision **Decides:** Adding steps and a formula removes the reasoning.

A	Correct	The revision does the thinking for the students.
B	Adjacent category	Formula use is not the same as higher-level thinking.
C	Confusion set	Demand depends on the thinking required, not the topic.
D	Inverted operation	More steps do not mean more reasoning.

9. Answer B Concept 6

A student in a class of 30 says the sum of two odd numbers is always odd. ~~The lesson is being recorded for a mentoring program.~~ Which teacher question is an advancing question that keeps the student reasoning?

Target word: choose a focusing/advancing question **Decides:** The question should push the student to test the claim.

A	Confusion set	A funneling question that leads to the answer.
B	Correct	Moves the student to test and revise their own thinking.
C	Adjacent category	An assessing question: it finds out thinking but does not advance it.
D	Inverted operation	Tells the answer and ends the reasoning.

10. Answer D Concept 6 · transfer

A teacher asks, “Is the slope positive? So it goes up? So the answer is B, right?” What kind of questioning is this?

Target word: classify the question sequence **Decides:** A chain of leading yes/no prompts.

A	Adjacent category	Wait time is a pause, not a question type.
B	Confusion set	These questions supply the thinking instead of eliciting it.
C	Inverted operation	Focusing questions leave the thinking to the student.
D	Correct	Each question narrows toward the teacher’s answer.

11. Answer D Concept 7

A new teacher ~~with a well-stocked supply closet~~ is planning a unit on exponential functions. ~~She has found a fun candy decay activity and a popular video.~~ Using backward design, what should she decide first?

Target word: first planning step **Decides:** Backward design starts from goals.

A	Dropped element	Evidence is step two; it must follow from the goals.
B	Camouflage	Starts from activities, the opposite of backward design.
C	Adjacent category	Coverage order is not a learning goal.
D	Correct	Goals come first, then evidence, then activities.

12. Answer B Concept 8

A student ~~who moved from another state in October~~ can set up a proportion but cannot solve it without help. ~~His family recently adopted a puppy.~~ What is the best support?

Target word: scaffold within reach **Decides:** He can do part of the task with support.

A	Absolute language	Scaffolds should fade as competence grows.
B	Correct	Support targets the step he cannot yet do alone and is removed as he gains skill.
C	Adjacent category	Changes the goal rather than supporting access to it.
D	Inverted operation	Doing the work removes the learning.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Conceptual before procedural	Procedures stick and transfer when they are connected to meaning; build the concept, then make the procedure efficient.	Assuming more drill fixes errors that come from missing understanding.	Students can perform a procedure but cannot explain or adapt it.
2	Formative assessment	Formative assessment gathers evidence during learning so the teacher can adjust instruction; its value lies in the adjustment, not the grade.	Treating any quiz as formative, or recording it as a grade and moving on.	Mid-unit evidence and “what should the teacher do next?”
3	Addressing misconceptions	A misconception is a consistent, reasoned error; surface it and create cognitive conflict (for example, test it with numbers) instead of just re-telling the rule.	Repeating the correct procedure louder, or treating the error as carelessness.	The same wrong pattern appears across many students, such as $(a + b)^2 = a^2 + b^2$.
4	Multiple representations	Moving between tables, graphs, equations, and words deepens understanding; the key is connecting the representations.	Adding representations side by side without linking them.	Students succeed in one form (the equation) but not another (the graph).
5	Productive struggle and cognitive demand	Keep the task’s thinking demand high and support the struggle with questions; do not turn the task into steps.	Rescuing students by giving the procedure, which lowers the demand.	Students are stuck on a rich task and the teacher must choose a move.
6	Assessing vs. advancing questions	Assessing questions find out what a student is thinking; advancing questions push the thinking forward. Focusing questions keep the student doing the reasoning; funneling questions lead to the answer.	Asking a chain of leading yes/no questions that funnel to the answer.	A dialogue in which the teacher chooses the next question.
7	Backward design	Plan from the learning goal, then decide the evidence of mastery, then choose activities.	Starting from favorite activities or the textbook page order.	“What should the teacher decide first?” when planning a unit.
8	Scaffolding within reach	Scaffolds target what a student can do with help but not yet alone, and are faded as competence grows.	Assigning easier, different content instead of supporting access to the same goal.	A student who can do part of a grade-level task with support.

Course scope (public description): Mathematics Learning and Teaching will help students develop the knowledge and skills necessary to become a prospective and practicing educator. This course will help students use a variety of instructional strategies to effectively facilitate the learning of mathematics. It focuses on selecting appropriate resources, using multiple strategies, and instructional planning, with methods based on research and problem solving. A deep understanding of the knowledge, skills, and disposition of mathematics pedagogy is necessary to become an effective secondary mathematics educator. There are no prerequisites for this course.